

Statistical Mechanics: Temperature from Entropy and Energy

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Chapter 1

Temperature from Entropy and Energy

Overview. Temperature first appears as a conversion of human thermometer units into microscopic energy units. After making the corresponding conversion from thermodynamic entropy to dimensionless statistical entropy, we describe equilibrium by a one-parameter family of probability distributions. The derivative dE/dS then defines temperature, and the first two laws show why a temperature difference determines the direction of energy flow.

1.1 Conversion Factors and Human Units

Many constants of physics are conversion factors. The speed of light converts a distance into a time:

$$t = \frac{x}{c}. \quad (1.1)$$

If distances and times are measured in compatible units, one may set $c = 1$. No physical effect disappears; the two kinds of measurement have simply been put on one common scale.

Boltzmann's constant has the same character. It converts the temperature read on a human-scale thermometer into an energy. The adjective *human-scale* is important. Such units are chosen because macroscopic instruments can produce and resolve them, not because microscopic dynamics singles them out.

The Celsius degree was fixed by dividing the interval between the freezing and boiling points of water at standard pressure into one hundred parts. Kelvin uses a degree of the same size but places zero at absolute zero. Choosing a different substance, a different interval, or Fahrenheit degrees would change the numbers but not the physics.

Question from the audience. Was Fahrenheit's scale already a seventeenth-century construction?

Response. The precise date is not needed here. The point is that all such scales were designed around temperatures that people could prepare and measure. Their degree size is conventional rather than microscopic. ¹

¹Lecture timestamp: 00:01:34–00:01:50.

At the time these scales were created, the molecular origin of heat was not available to experiment. A one-degree change could be observed with a macroscopic thermometer, whereas the energy of one molecule could not be measured. It was therefore natural to invent a temperature unit before anyone knew that temperature is fundamentally an energy scale.

1.2 Why Temperature Has Units of Energy

Consider a dilute monatomic gas. Dilute means that interaction energies between molecules may be neglected for the present purpose. In thermal equilibrium, equipartition assigns $T_K/2$ times Boltzmann's constant to each quadratic translational degree of freedom. In three spatial dimensions,

$$\langle E_{\text{kin}} \rangle = \frac{3}{2} k_B T_K. \quad (1.2)$$

The factor three counts motion in the x , y , and z directions. The factor one-half comes from the quadratic kinetic energy.

This relation is not peculiar to one molecular species. In a mixture of helium and oxygen at equilibrium, every freely translating particle has the same average center-of-mass kinetic energy, even though the heavier particles move more slowly. Since

$$E_{\text{kin}} = \frac{1}{2} m v^2, \quad (1.3)$$

equal characteristic energy implies $v \propto m^{-1/2}$.

The same idea can be pushed to a deliberately extreme example. Suspend a bowling ball in a fluid and let the complete system reach equilibrium. Its Brownian translational degrees of freedom are governed by the same energy scale as those of a molecule. The ball's velocity is fantastically smaller because its mass is fantastically larger, but temperature compares energy scales, not speeds.

At room temperature, T_K is about 300 K, while the energy of one microscopic degree of freedom is tiny in joules. Consequently the conversion factor is tiny:

$$k_B \simeq 1.38 \times 10^{-23} \text{ J K}^{-1}. \quad (1.4)$$

Its scale and Avogadro's number are related rather than coincidental:

$$R = N_A k_B. \quad (1.5)$$

The gas constant R describes a mole, whereas k_B describes the corresponding scale per microscopic constituent.

Fundamental formulas repeatedly contain the combination $k_B T_K$. It is therefore cleaner to define

$$T \equiv k_B T_K. \quad (1.6)$$

From now on T itself has units of energy, and (1.2) becomes

$$\langle E_{\text{kin}} \rangle = \frac{3}{2} T. \quad (1.7)$$

If energy is measured in joules, temperature is measured in joules. If energy is measured in electron volts, temperature is measured in electron volts. There is no new unit of nature called a temperature unit.

1.3 Two Conventions for Entropy

The same conversion appears on the entropy side. Macroscopic thermodynamics grew out of the analysis of heat engines. Carnot supplied the decisive engine reasoning; Clausius later introduced the thermodynamic entropy and the symbol S . In the notation used here, call that macroscopic quantity S_C . Its units are energy divided by thermometer temperature, for example joules per kelvin.

Statistical entropy is dimensionless. For a discrete distribution,

$$S = - \sum_i p_i \log p_i. \quad (1.8)$$

With natural logarithms a fair binary alternative has entropy $\log 2$; with base-two logarithms the same amount is called one bit. The two entropy conventions are related by

$$S = \frac{S_C}{k_B}, \quad S_C = k_B S. \quad (1.9)$$

A macroscopic value modest in J K^{-1} corresponds to an enormous dimensionless entropy because it counts information distributed among an enormous number of microscopic constituents.

Equations (1.6) and (1.9) establish the convention used throughout:

$$[T] = [\text{energy}], \quad [S] = 1.$$

Boltzmann's constant may now disappear from the working formulas. It can always be restored by replacing T with $k_B T_K$ and S with S_C/k_B .

Question from the audience. Apart from the factor $3/2$, is the new temperature simply an energy measured in joules?

Response. Yes, if joules are the chosen energy unit. A joule is the MKS unit $\text{kg m}^2 \text{s}^{-2}$. One could instead use electron volts or any other energy unit; temperature must then be expressed in that same unit. The factor $3/2$ belongs to the ideal-gas example, not to the definition of temperature. ²

Question from the audience. If the board specifies kelvin, is k_B not then a definite constant?

Response. It is definite after both the temperature and energy units have been specified. Change kelvin to Fahrenheit degrees, or joules to electron volts, and its numerical value changes. That unit dependence is exactly the behavior of a conversion factor. ³

Question from the audience. Is the numerical value of Boltzmann's constant determined experimentally?

Response. Yes. Historically its evaluation required microscopic information about atoms and their number density. Boltzmann understood the relation between molecular energy and temperature before those quantities were known precisely, much as Newton knew the role of the gravitational constant before it had been measured accurately. ⁴

²Lecture timestamp: 00:15:49–00:16:45.

³Lecture timestamp: 00:16:46–00:18:33.

⁴Lecture timestamp: 00:18:46–00:20:00.

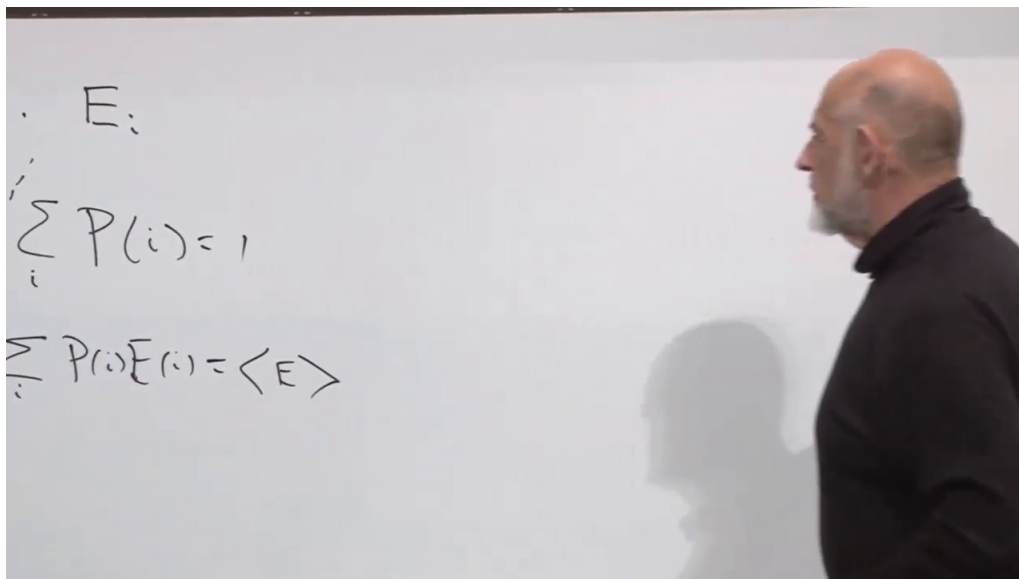


Figure 1.1: Normalization and average energy on the board: $\sum_i P(i) = 1$ and $\sum_i P(i) E(i) = \langle E \rangle$.

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1.4 Equilibrium Is a Statistical State

The unit conversion still has not defined temperature. Touching something hot gives a useful biological intuition, but temperature is a derived physical quantity built from energy and entropy.

Imagine a system in contact with a large environment whose macroscopic conditions are fixed. Energy can pass between them. If the system initially has too much energy, it loses energy on average; if it has too little, it gains energy on average. In thermal equilibrium microscopic exchanges continue, but there is no net average flow in either direction.

Let i label an exact microscopic state of the system and let E_i be that state's energy. Equilibrium does not mean that the system freezes in one state. Its interaction with the environment continually changes the microstate, so we assign a probability P_i to each possibility. The basic relations are

$$\sum_i P_i = 1, \quad (1.10)$$

$$\sum_i P_i E_i = \bar{E}, \quad (1.11)$$

$$S = - \sum_i P_i \log P_i. \quad (1.12)$$

The bar on $\bar{E}$ emphasizes that the subsystem's energy fluctuates. One measurement may give one value and the next another; repeated measurements settle into the distribution whose mean is (1.11).

1.5 A Family Labeled by Mean Energy

The equilibrium distribution depends strongly on the mean energy. Put a pot of water on a stove and $\bar{E}$ rises; put it in the snow and $\bar{E}$ falls. The probabilities change with it. Equilibrium should

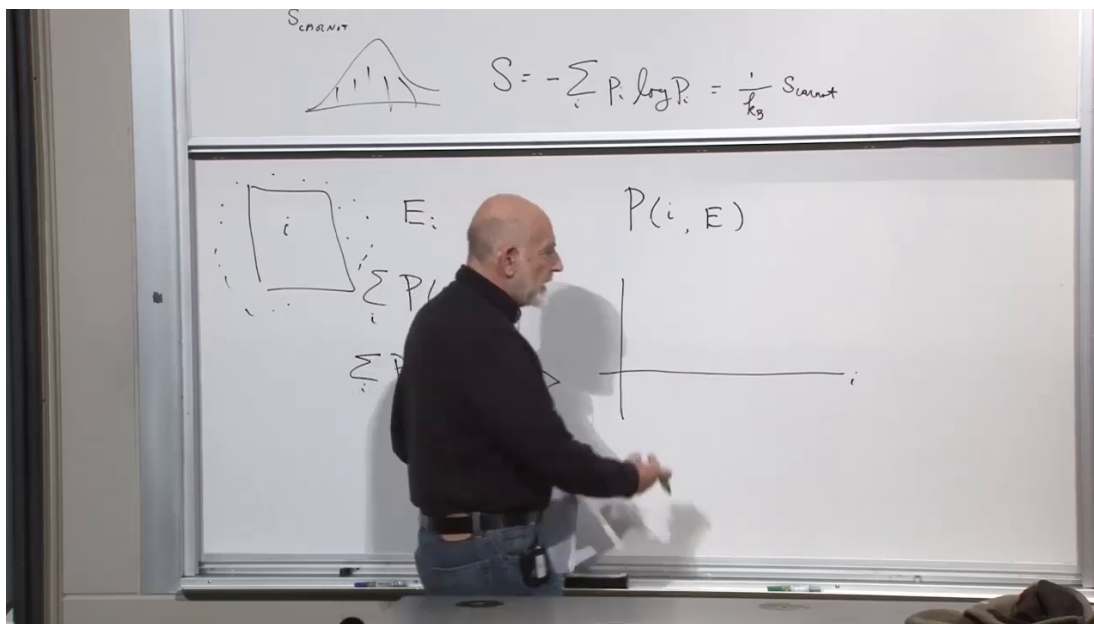


Figure 1.2: The transition to $P(i, \bar{E})$. The earlier normalization, mean-energy, and entropy equations remain visible while the family of distributions is introduced.

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therefore be regarded not as one distribution but as a one-parameter family:

$$P_i(\bar{E}). \quad (1.13)$$

Every member satisfies

$$\sum_i P_i(\bar{E}) = 1, \quad (1.14)$$

$$\sum_i P_i(\bar{E}) E_i = \bar{E}. \quad (1.15)$$

For a schematic plot, order the states by increasing E_i . The horizontal axis may then be read either as the index i or as the corresponding energy. Higher mean energy shifts probability weight toward states farther to the right. Every curve must retain unit total area.

Question from the audience. Is the notation circular? The mean energy is computed from the distribution, while the distribution is labeled by the mean energy.

Response. Begin with an empirically selected one-parameter family of equilibrium distributions. Each member has a mean energy. If distinct members have distinct means, that mean may be used as the parameter naming the family. The notation does not yet derive the family; the Boltzmann distribution will provide that derivation later. ⁵

⁵Lecture timestamp: 00:28:35–00:29:44.

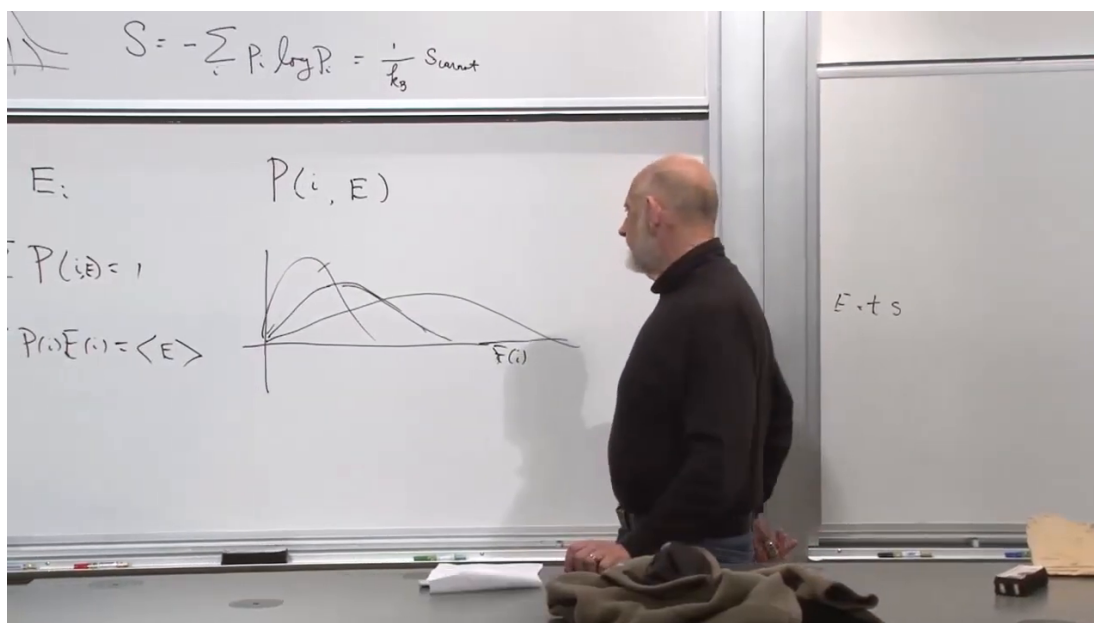


Figure 1.3: Several schematic distributions $P(i, \bar{E})$. Curves weighted farther toward the right have larger mean energy, while each remains normalized.

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Question from the audience. Why say average energy rather than total energy?

Response. The subsystem is exchanging energy with a heat bath, such as the atmosphere around a gas container with thermally permeable walls. Its instantaneous energy fluctuates. Many measurements produce a distribution, and $\bar{E}$ is the mean of those outcomes. The energy of the larger closed system consisting of subsystem plus bath may still be fixed. ⁶

Question from the audience. Can different microscopic states have the same energy, and can different distributions have the same mean energy?

Response. Both are possible. The mean-energy constraint by itself allows many distributions, so the special one-parameter equilibrium family requires an additional physical argument. Its origin and its Boltzmann form will be derived later. Equal energies of distinct microstates are the separate issue called degeneracy. ⁷

1.5.1 State labels, degeneracy, and correlation

Several distinctions that are easy to blur must be kept separate.

First, many probability distributions can have the same mean energy. The one-parameter equilibrium family is not obtained from the mean-energy constraint alone. It is the particular family selected by equilibration; its form will later be identified as the Boltzmann distribution.

⁶Lecture timestamp: 00:29:46–00:30:56.

⁷Lecture timestamp: 00:30:58–00:32:22.

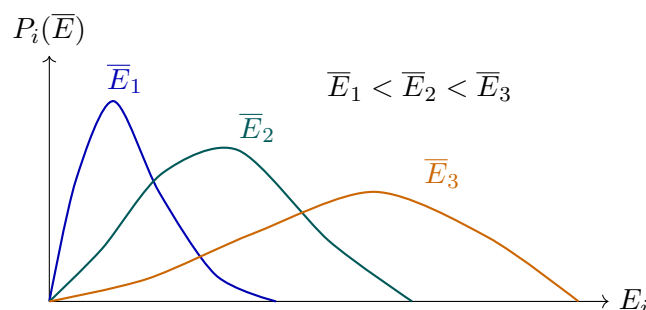


Figure 1.4: A clean version of the board sketch. The curves need not have these precise shapes; only normalization and their ordered mean energies are being used.

Second, several different microstates can have the same energy. This is called degeneracy. The board discussion temporarily ignores exact degeneracies so that the states can be ordered unambiguously by energy. Degeneracy is in fact common when a system has symmetries, but it causes no conceptual problem: retain the state label i , allow repeated values of E_i , and sum over all states.

Third, correlation is not another word for degeneracy. Correlation concerns a joint distribution of two random quantities and asks whether knowing one changes the probabilities for the other. The mutually exclusive labels i and j in one list of microstates are not, by themselves, two random variables whose correlation is being computed.

Question from the audience. Does the construction assume that different i -states are independent, and what if two of them are correlated?

Response. A state label is one possible value of a single microscopic random variable. Independence and correlation require two jointly distributed variables, so that terminology does not apply directly to two alternative labels in this list. Equal energies of distinct states are instead a question about degeneracy. ⁸

Question from the audience. What if two distinct states have exactly the same energy?

Response. That is degeneracy. For the immediate graph we adopt the simplifying assumption that no two energies coincide and order the states by increasing energy. The later probability formulas do not require this assumption; degenerate states are simply counted separately. ⁹

Question from the audience. Is the meaning of the index i arbitrary?

Response. The microscopic states themselves are not arbitrary: each specifies all positions, momenta, or other data needed for an exact description. Their names are arbitrary, so for the picture we choose labels in ascending order of energy. ¹⁰

⁸Lecture timestamp: 00:32:23–00:33:22.

⁹Lecture timestamp: 00:33:23–00:34:04.

¹⁰Lecture timestamp: 00:34:06–00:35:15.

1.6 Entropy Becomes a Function of Energy

Every member of the equilibrium family has an entropy. Substituting (1.13) into (1.12) gives

$$S(\bar{E}) = - \sum_i P_i(\bar{E}) \log P_i(\bar{E}). \quad (1.16)$$

This is not one universal function shared by all physical systems. It depends on the system's spectrum and on its equilibrium family. Once that family is specified, however, energy and entropy are linked: moving through the family changes both.

Ordinary systems typically spread their probability over more states as their mean energy rises, so S usually rises with $\bar{E}$. For now we assume that this relation is smooth and locally invertible. The exceptions are important, but the simple case exposes the definition we need.

1.7 The Derivative That Defines Temperature

Ask a concrete question: how much must the mean energy change in order to increase the entropy by one binary alternative? In natural-log units that entropy increment is $\log 2$.

For neighboring members of a smooth family,

$$dE = \frac{dE}{dS} dS. \quad (1.17)$$

At this point the equation is only the chain rule. Its coefficient, however, has exactly the dimensions and physical role required of temperature.

Definition 1.1. The temperature in energy units is

$$T \equiv \frac{dE}{dS}. \quad (1.18)$$

Equivalently,

$$dE = T dS, \quad dS = \frac{1}{T} dE. \quad (1.19)$$

For a sufficiently small change of one bit,

$$\Delta E \simeq T \log 2. \quad (1.20)$$

The fixation on one bit is only a way to make the derivative tangible. Any small entropy change obeys the same local relation.

Question from the audience. Since entropy was written as $S(E)$, would dS/dE not be the more natural derivative?

Response. Either derivative contains the same information when the relation is locally invertible:

$$\frac{dE}{dS} = \left(\frac{dS}{dE} \right)^{-1}.$$

Temperature is conventionally dE/dS , while inverse temperature is $dS/dE = 1/T$.¹¹

¹¹Lecture timestamp: 00:40:24–00:40:54.

Only one thermodynamic variable is changing here, so ordinary derivatives are sufficient. When volume, particle number, magnetization, or other variables are allowed to change, one must use partial derivatives and state which quantities are held fixed. The present definition is the one-variable slice of that more general thermodynamic geometry.

1.7.1 Why Boltzmann's constant cancels

Return briefly to thermometer units. Using

$$T = k_B T_K, \quad S = \frac{S_C}{k_B},$$

the fundamental relation becomes

$$dE = (k_B T_K) d\left(\frac{S_C}{k_B}\right) \quad (1.21)$$

$$= T_K dS_C. \quad (1.22)$$

One factor of k_B enters through temperature and the reciprocal factor enters through entropy. They cancel. This is why a conventional thermodynamics text can write (1.22) without displaying Boltzmann's constant.

Question from the audience. Must $S(E)$ be invertible?

Response. Not always. For ordinary systems in the regime considered here, increasing energy generally broadens the equilibrium distribution and increases entropy, so $S(E)$ is monotone and $T > 0$. Systems with bounded spectra can provide counterexamples, including regimes in which entropy decreases with energy and the statistical temperature is negative. We set those cases aside for now. ¹²

1.8 From the Abstract Derivative to Hot and Cold

The derivative definition is universal but still abstract. We want to recover the familiar statement that energy flows from a hotter body to a colder one. Pain from touching a hot surface is biological; color and emitted radiation depend on the properties of light. A statistical definition should also make sense for an idealized gas even when radiation is ignored.

Question from the audience. Could the light emitted by a hot object provide the basic definition of temperature?

Response. Radiation is a useful thermometer, but it introduces the physics of light. Statistical mechanics must define temperature for systems such as a box of gas without relying on optical properties. The direction of energy exchange is the more general criterion. ¹³

Prepare two systems A and B , each initially in its own equilibrium, with temperatures T_A and T_B . They are initially isolated from one another. Then connect them through a weak thermal contact:

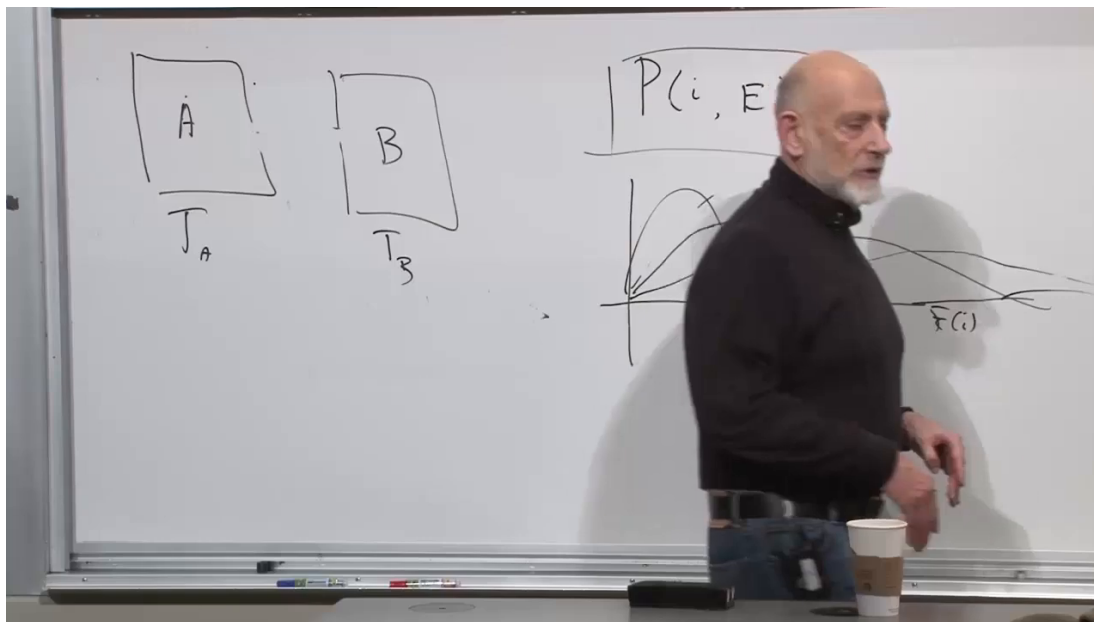


Figure 1.5: The separately prepared systems A and B , labeled by temperatures T_A and T_B , just before weak thermal contact is considered.

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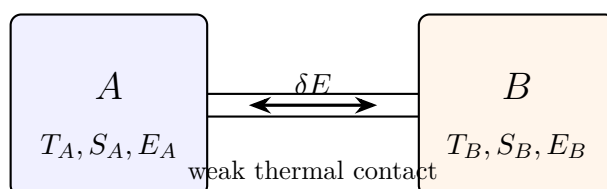


Figure 1.6: Energy fluctuates through the weak contact while the combined system remains isolated.

the *thin pipe* allows small amounts of energy to pass back and forth without destroying the identity of either subsystem.

Suppose, without loss of generality,

$$T_B > T_A. \quad (1.23)$$

Experience says that B should lose energy and A should gain it. Three facts turn that expectation into a calculation.

1.8.1 Entropy additivity

For two weakly coupled subsystems whose microscopic alternatives combine independently, the number of joint possibilities multiplies:

$$W_{AB} = W_A W_B.$$

Entropy is logarithmic, so

$$S_{\text{tot}} = S_A + S_B. \quad (1.24)$$

¹²Lecture timestamp: 00:44:30–00:45:57.

¹³Lecture timestamp: 00:46:42–00:47:20.

The same result follows directly from the Shannon entropy of a product distribution.

Question for the class. Why does the entropy of the combined system add?

Response. Because the underlying number of alternatives multiplies while entropy takes a logarithm: $\log(W_A W_B) = \log W_A + \log W_B$. Weak interaction is what permits the subsystem description and this leading additive form. ¹⁴

1.8.2 The first and second laws

The combined system is isolated, so energy exchanged internally is conserved:

$$dE_A + dE_B = 0, \quad dE_B = -dE_A. \quad (1.25)$$

This is the first law in the present experiment.

The second law is taken here as a postulate to be explained more deeply later. As the connected systems relax toward their new equilibrium,

$$dS_{\text{tot}} = dS_A + dS_B \geq 0. \quad (1.26)$$

The strict inequality describes spontaneous relaxation away from equilibrium; equality holds at equilibrium for the reversible first-order variation.

Finally, the temperature definition applies to each subsystem:

$$dE_A = T_A dS_A, \quad dE_B = T_B dS_B. \quad (1.27)$$

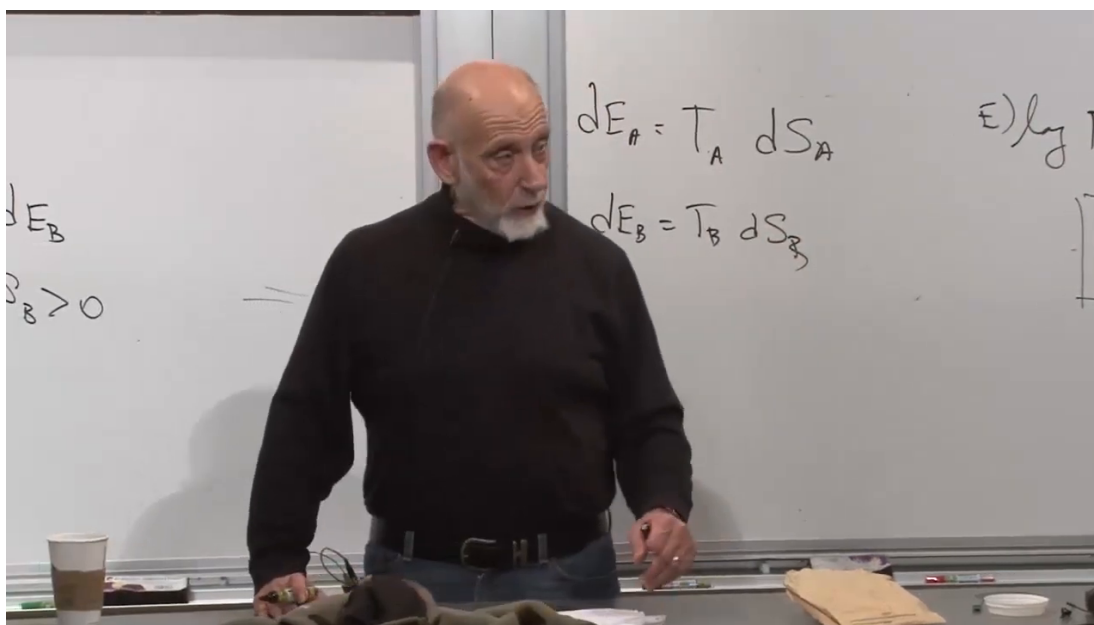


Figure 1.7: The two subsystem relations $dE_A = T_A dS_A$ and $dE_B = T_B dS_B$ at the close of the board argument.

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¹⁴Lecture timestamp: 00:51:06–00:52:03.

1.8.3 The sign of the energy flow

These equations already fix the direction of net transfer. Substitute $dS_A = dE_A/T_A$, $dS_B = dE_B/T_B$, and (1.25) into (1.26):

$$dS_{\text{tot}} = \frac{dE_A}{T_A} + \frac{dE_B}{T_B} \quad (1.28)$$

$$= dE_A \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \geq 0. \quad (1.29)$$

If $T_B > T_A > 0$, then

$$\frac{1}{T_A} - \frac{1}{T_B} > 0.$$

Equation (1.29) therefore requires

$$dE_A > 0, \quad dE_B < 0.$$

The colder system A gains energy and the hotter system B loses it. When $T_A = T_B$, the first-order entropy change supplies no preferred direction, and microscopic exchanges balance to zero net flow. Thus the abstract derivative $T = dE/dS$ reproduces the operational distinction between hot and cold.